

ECL-257 Integrated Circuit Design

Project Report: 4-bit synchronous Up Counter

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Introduction

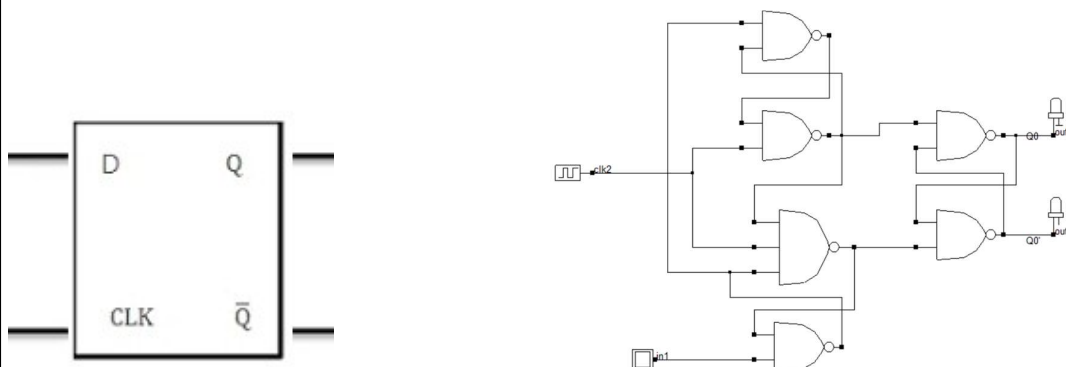
Counters are key sequential circuits in VLSI, made using cascaded flip-flops. They are widely used for counting, timing, and frequency measurement in devices like clocks and automotive systems. Based on how the clock is applied, counters are classified as asynchronous or synchronous. In synchronous counters, all flip-flops share the same clock source, eliminating propagation delay and supporting higher operating frequencies. A 4-bit synchronous up-counter (using J-K flip-flops) counts from 0000 to 1111 and then resets. Since DSCH cannot directly create J-K flip-flops, they are implemented using D flip-flops, after which the 4-bit synchronous counter is built.

Components Used and Specifications:

Since this project have not been implemented practically, therefore the project was entirely implemented in DSCH v3 and MICROWIND v3 software. The list of materials/components that have been used during implementing this project are given below:

STRUCTURE	NAME OF MATERIAL
D Flip flop	NAND GATE-5 , CLOCK PULSE
J-K FLIP FLOP	NOT GATE -1, OR GATE-1 , AND GATE-2 , D FLIP FLOP -1, LED OUTPUT-2 , CLOCK PULSE, J-K INPUT
4- BIT SYNCHRONOUS UP COUNTER	J-K FLIP FLOP-4 , AND GATE -2 , LED OUTPUT-4, VOLTAGE SOURCE-INPUT, CLOCK PULSE GENERATOR

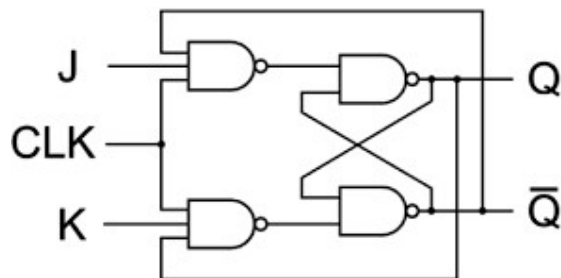
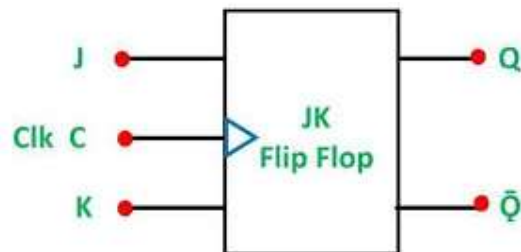
Working Principle : A D flip-flop is derived from an SR flip-flop by adding a NOT gate so that it uses a single "Data (D)" input instead of separate Set and Reset inputs. When D = 1, the flip-flop sets; when D = 0, it resets. To avoid unwanted changes on every input pulse, a clock (or enable) input is added. The output Q updates only when the clock is active; when the clock goes low, the output is latched and remains unchanged. This makes the D flip-flop a stable, edge-controlled storage element used in synchronous circuits.



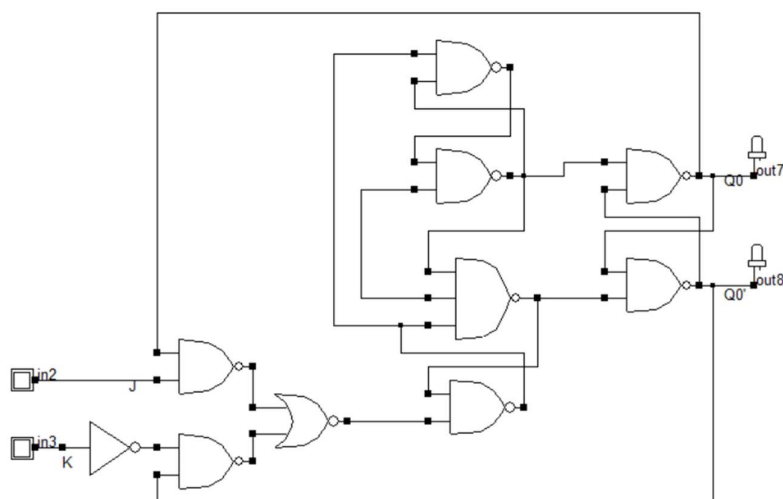
J-K FLIP FLOPS: The J-K flip-flop is the most commonly used one among the all basic flip-flops. It has the input-following the character of the clocked D flip-flop but has two external inputs, labelled as J and K. If J and K mismatches with each other only then the value of J will be taken by the output Q at the next clock edge. The symbol, circuit diagram and the truth table of J-K Flop Flops are given below.

Truth Table

J	K	CLK	Q
0	0	↑	Q_0 (no change)
1	0	↑	1
0	1	↑	0
1	1	↑	$\bar{Q}_0$ (toggles)

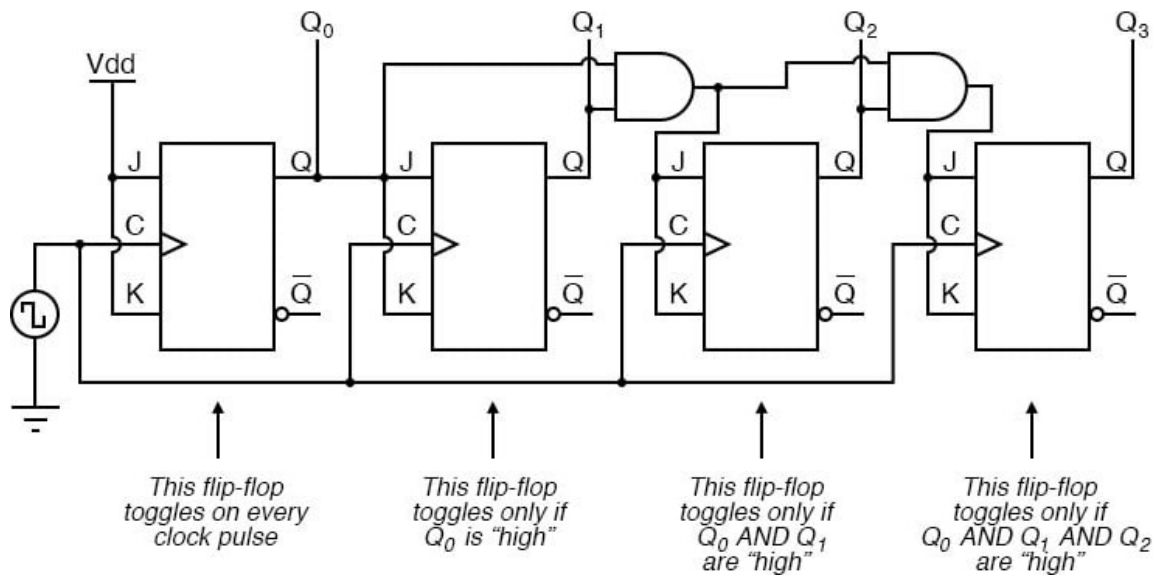


Jk flip flop made using d-flip flop: A J-K flip-flop can be created using a D flip-flop by generating the required D input from the external J and K signals. Since DSCH cannot directly implement a J-K flip-flop, the D flip-flop was first designed and then used to build the J-K flip-flop through additional logic gates.



4-Bit Synchronous Up Counter (using J-K Flip-Flops):

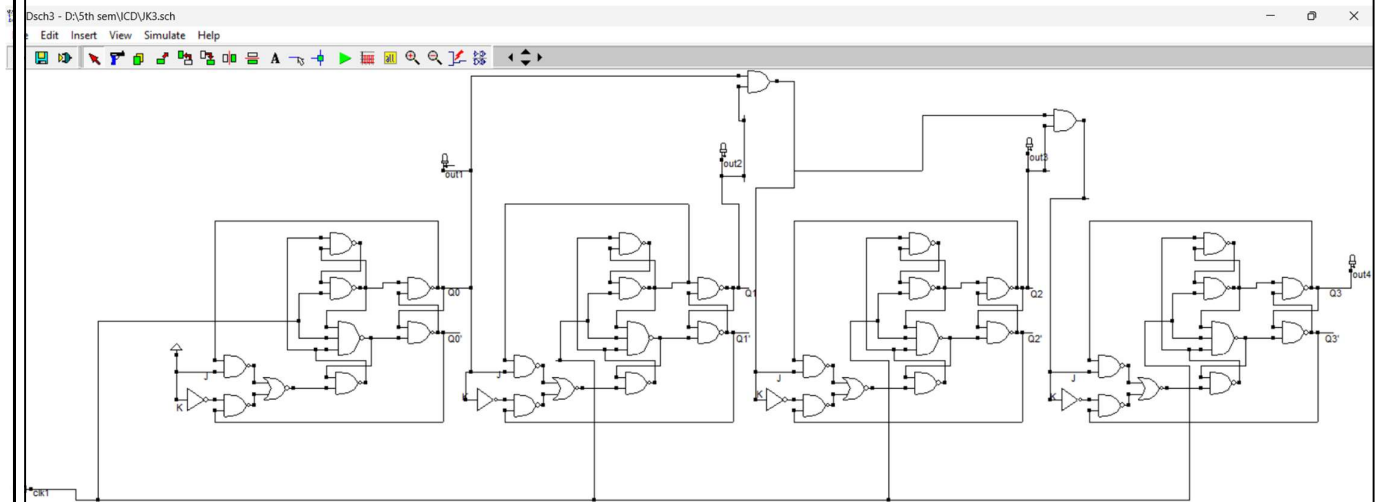
A 4-bit synchronous up counter uses four J-K flip-flops, all driven by the same clock. The LSB flip-flop toggles on every clock pulse since its J and K inputs are tied high. Each higher-order flip-flop toggles only when all the previous Q outputs are high, which is achieved using AND gates. This ensures a proper binary sequence from 0000 to 1111. The entire counter was then implemented and tested in DSCH.



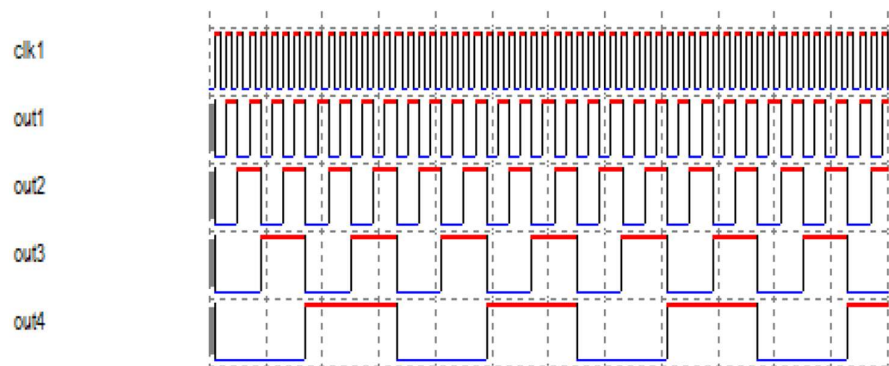
Truth table of a 4-Bit Synchronous Up Counter:

Q ₃	Q ₂	Q ₁	Q ₀
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1

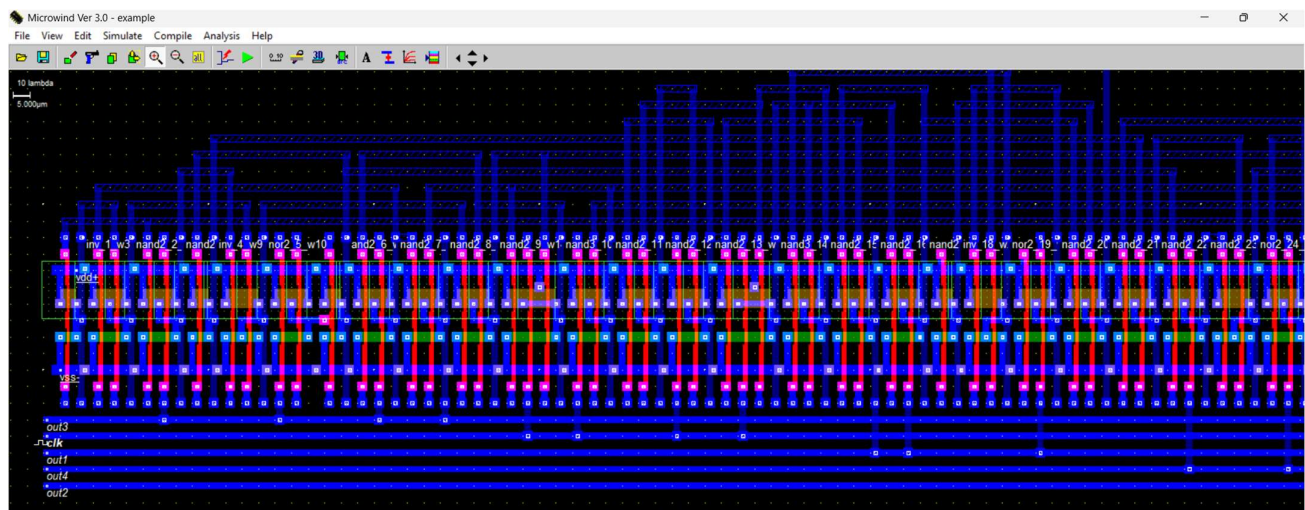
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1



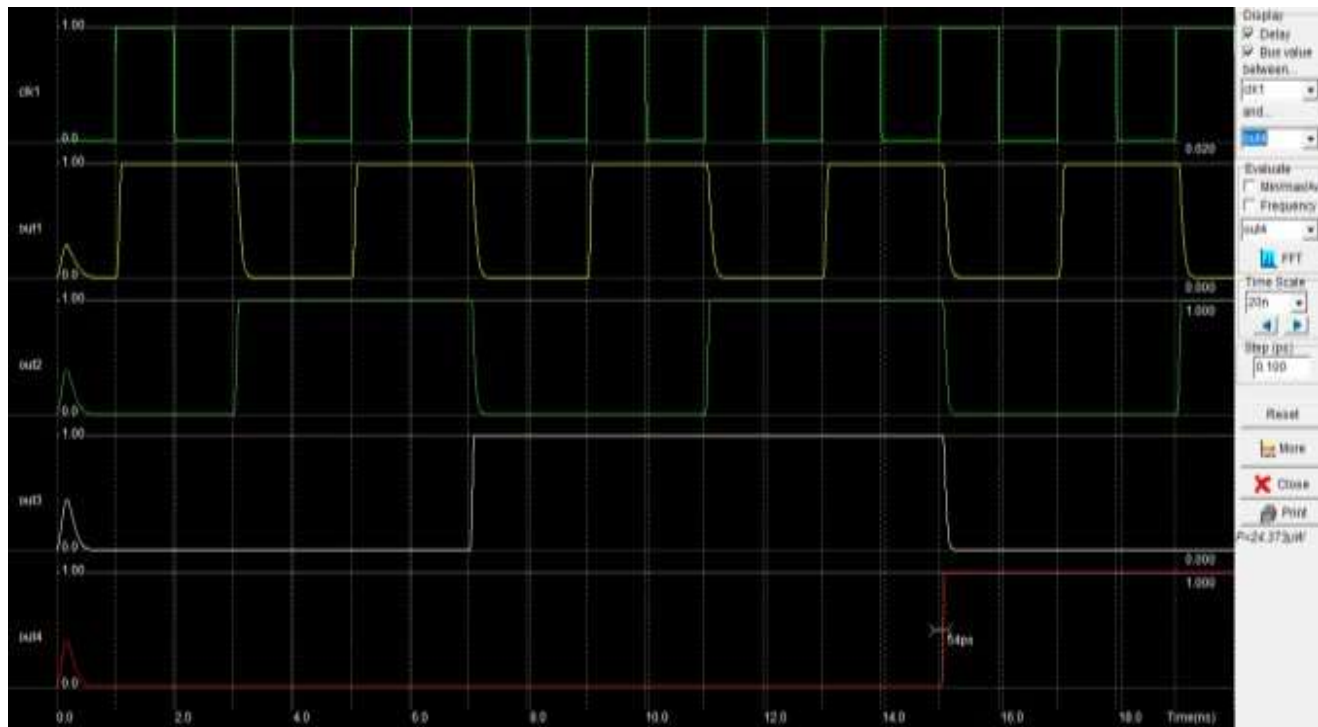
RESULTS: IN DSCH



Result in microwind



Result in Microwind : Clock and output graphs



Conclusion: This paper presented the design of a 4-bit synchronous up counter using DSCH v3. First, a D flip-flop was created, which was then used to build a J-K flip-flop using additional logic gates. After verifying both circuits through timing diagrams, four J-K flip-flops were connected to form the synchronous counter. The counter successfully counted from 0000 to 1111 with a common clock input. Initial timing errors were corrected through debugging, and the final output matched the expected truth table. Although synchronous counters may face complexity and clock-skew issues as bit-width increases, the simulated results here were accurate and consistent with theory.

